

BOARD OF INTERMEDIATE EDUCATION KARACHI

HSC PART I · CLASS XI

Computer Science Lecture Notes

Worked examples, board-style practice, and practicals

Session 2026–27

Sindh Textbook Board chapter scheme

New Sindh Curriculum of Computer Science 2024

Aligned with the National Curriculum of Pakistan 2022–23

Theory 75 marks · Practical 25 marks

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How to use these lectures

These notes are for classroom teaching of Computer Science at the Board of Intermediate Education Karachi. They follow the six units of the Sindh Textbook Board Class XI book for the session that uses the New Sindh Curriculum of Computer Science 2024, aligned with the National Curriculum of Pakistan 2022–23. They are original teaching notes. They are not an official BIEK paper, not a Sindh Textbook Board book, and not a replacement for the prescribed textbook or the current model paper.

How the book is arranged

Each lecture states what a student should be able to do, teaches the idea with a worked example, and then gives board-style questions with model answers. Read the example before the answer key. In class, cover the model answer until the attempt is on paper. The practicals match the 25 practical marks: keep a journal with the aim, the steps, the output, and one fault you fixed.

Lecture	Textbook unit	You will be asked to
1	Computer Systems	Use gates and truth tables, describe the SDLC, and compare topologies and basic security
2	Computational Thinking and Algorithms	Write and trace pseudocode, linear search, binary search, bubble sort, and insertion sort
3	Programming Fundamentals	Write Python with decisions, loops, turtle graphics, lists, and functions
4	Data and Analysis	Model a small database and describe data without calling an association a cause
5	Applications and Impacts	Explain IoT, cloud, and blockchain in a real setting, and judge a source
6	Digital Literacy	Search, collect, and present data, and test a small prototype

The Class XI book, and the national domains those units carry.

Marks

In the BIEK scheme of studies, Computer Science Paper I is 75 marks of theory and 25 marks of practical. Recent higher-secondary patterns split theory into multiple choice, short answers, and long answers, often near 20, 40, and 40 percent. Confirm the model paper for the year you will sit. These notes practise all three kinds. A practical journal that only contains printed output, with no input and no fault, is a weak practical.

How to write an answer

- Start a short answer with the definition the question named, then one precise point or a tiny example. Stop.
- Start a long answer with a one-line claim, then headings. A table or a trace on its own line is easier to mark than a paragraph that hides the numbers.
- Use the scenario in the question. A memorised definition that never mentions the college, the list, or the sensor loses the application marks.
- If you are unsure of a figure, show the working. A wrong final line with a correct method can still earn method marks. An invented p-value cannot.
- For a program, write the sample input and the output your code would actually produce.

A LIMIT ON SECURITY TOPICS

Where the curriculum names an attack, these notes say what harm it causes and what a student or a college should do instead. They do not give instructions for breaking into a system, building malware, or sending phishing messages. That work is not the practical.

CLASS XI · UNIT 1

Lecture 1

Computer Systems

Textbook unit. Unit 1 — Computer Systems

Curriculum domain. A. Computer systems

Class time. about 30 periods

A computer system is not only the box on the desk. It is hardware, software, data, and the rules that move that data safely from one place to another. This lecture follows the Class XI textbook unit: how analog and digital signals differ, how Boolean logic becomes a circuit, how a software project moves from an idea to a tested product, and how networks are organised so they can grow and keep working.

Learning outcomes

- Distinguish analog and digital signals and build truth tables for up to three inputs.
- Write Boolean expressions, apply the basic identities and duality, and draw the matching logic diagram.
- Relate the stages of the software development life cycle to a small case, and compare Waterfall with Agile.
- Explain black-box and white-box testing.
- Describe the basic communication model, the OSI layers, and the TCP/IP model.
- Compare network topologies for scalability and reliability.
- Explain why cybersecurity matters and contrast encryption with hashing, at the level of purpose and use.

Analog and digital representation

An **analog** signal changes smoothly. A microphone turns a voice into a voltage that rises and falls with the sound. A **digital** signal uses a fixed set of values. Inside a computer those values are two voltage bands that we call 0 and 1. A **bit** is one such value. Eight bits make a **byte**.

Digital representation is the reason a file can be copied without slowly becoming blurry. Each bit is restored to a clean 0 or 1 at every step. The cost is that a smooth real-world signal must be **sampled** (measured at moments in time) and **quantised** (rounded to the nearest allowed level). A higher sampling rate and more bits per sample keep more detail, and they also make a larger file.

DIGITAL LOGIC

Digital logic is the set of rules that combine bits. Those rules are written as Boolean expressions and built in hardware as logic gates. Every later topic in this lecture — circuits, programs, and even encryption — rests on bits being cleanly 0 or 1.

Boolean expressions, functions, and identities

A **Boolean variable** is a name that stands for 0 or 1. We write 0 for false and 1 for true. A **Boolean expression** combines variables with operators. A **Boolean function** is the rule that maps each possible input combination to one output. The truth table is the full listing of that function.

In this course the operators are written in words so the paper stays readable: **AND** (also written as a dot), **OR** (also written as +), and **NOT** (a bar or a prime, so the NOT of A is A'). AND gives 1 only when every input is 1. OR gives 1 when at least one input is 1. NOT flips its single input.

Name	OR form	AND form
Identity	$A + 0 = A$	$A \cdot 1 = A$
Null (dominance)	$A + 1 = 1$	$A \cdot 0 = 0$

Name	OR form	AND form
Idempotent	$A + A = A$	$A \cdot A = A$
Complement	$A + A' = 1$	$A \cdot A' = 0$
Double negation	$(A')' = A$	$(A')' = A$
Commutative	$A + B = B + A$	$A \cdot B = B \cdot A$
Associative	$(A + B) + C = A + (B + C)$	$(A \cdot B) \cdot C = A \cdot (B \cdot C)$
Distributive	$A + (B \cdot C) = (A + B) \cdot (A + C)$	$A \cdot (B + C) = (A \cdot B) + (A \cdot C)$
Absorption	$A + (A \cdot B) = A$	$A \cdot (A + B) = A$
De Morgan	$(A + B)' = A' \cdot B'$	$(A \cdot B)' = A' + B'$

Table 1. Boolean identities used in Class XI. A and B are inputs. A' is NOT A.

Duality swaps the operators and the constants: every OR becomes AND, every AND becomes OR, every 0 becomes 1, and every 1 becomes 0. Variables are left as they are. The dual of a true identity is another true identity. The dual of $A + 0 = A$ is $A \cdot 1 = A$. The dual of $A + A' = 1$ is $A \cdot A' = 0$. Duality does not mean the dual expression has the same output as the original. It means the dual *law* is also a law.

WORKED EXAMPLE — SIMPLIFY AND CHECK

Simplify $F = A \cdot (A + B)$.

By the absorption law, $A \cdot (A + B) = A$. So the circuit ignores B.

Check one row. If $A = 1$ and $B = 0$, then $A + B = 1$, and $A \cdot 1 = 1$, which equals A. If $A = 0$ and $B = 1$, then $A + B = 1$, and $0 \cdot 1 = 0$, which equals A. The identity holds.

WORKED EXAMPLE — DE MORGAN

Show that $(A + B)'$ and $A' \cdot B'$ match for $A = 0$, $B = 1$.

$A + B = 0 + 1 = 1$, so $(A + B)' = 0$.

$A' = 1$ and $B' = 0$, so $A' \cdot B' = 0$. Both sides agree. A full proof is the four-row truth table; one matching row is only a check, not a proof.

Logic gates and truth tables

A **logic gate** is the hardware for one operator. A **switch** is an older picture of the same idea: two switches in series act as AND (both must be closed), and two switches in parallel act as OR (either one is enough). You may be asked to recognise a gate from its shape or from its table. Learn both the two-input table and the name of the gate that inverts it.

A	B	AND	OR	NAND	NOR	XOR	XNOR
0	0	0	0	1	1	0	1
0	1	0	1	1	0	1	0
1	0	0	1	1	0	1	0
1	1	1	1	0	0	0	1

Table 2. Two-input gates. NAND is NOT of AND. NOR is NOT of OR. XOR is 1 when the inputs differ. XNOR is 1 when they match.

NOT has one input: 0 becomes 1, and 1 becomes 0. **NAND** and **NOR** are called universal gates because any other gate can be built from NAND alone, or from NOR alone. **XOR** is the odd-one-out detector: $A \text{ XOR } B = A' \cdot B + A \cdot B'$.

For three inputs, list the rows in binary counting order from 000 to 111 so you never skip a row. There are $2^3 = 8$ rows. Compute the expression in small pieces, column by column, instead of doing it in your head.

WORKED EXAMPLE — THREE-INPUT TRUTH TABLE

Build the truth table for $F = A \cdot B + C'$.

Order the rows as ABC from 000 to 111. First column: $A \cdot B$ is 1 only on the last two rows (110 and 111). Second column: C' is 1 whenever C is 0, that is on rows 000, 010, 100, and 110. F is 1 where either column is 1, so $F = 1$ on 000, 010, 100, 110, and 111. $F = 0$ on 001, 011, and 101.

To **draw a logic diagram** from an expression, start inside the brackets. For $F = (A \text{ AND } B) \text{ OR } (\text{NOT } c)$ draw an AND gate for A and B, a NOT gate for C, and feed both results into an OR gate. To **read an expression from a diagram**, label the wire after every gate and combine those labels at the next gate. The final wire is F.

A first look at Karnaugh maps

A **Karnaugh map** (K-map) is a truth table redrawn so that neighbouring cells differ by only one variable. Grouping neighbouring 1s lets you drop the variable that changes inside the group. Groups must be rectangles of 1, 2, 4, or 8 cells. This course expects you to recognise the map and simplify a small one, not to handle five variables.

WORKED EXAMPLE — A MAP THAT COLLAPSES TO ONE VARIABLE

Let the minterms where $F = 1$ be the four cases in which $B = 0$: ABC = 000, 001, 100, and 101. Those four cells form one group. Inside the group, A takes both values and C takes both values, so A and C are not needed. B is 0 throughout, which is written B' . So $F = B'$.

Check: when B is 0, F is 1 no matter what A and C are. When B is 1, F is 0. That is exactly B' .

The software development life cycle

The **software development life cycle (SDLC)** is the planned path from a need to a working system that somebody can maintain. A **bug** is a fault that makes the system do the wrong thing. **Debugging** is the work of finding and removing that fault. The engineering point of the SDLC is that quality and risk are managed on purpose, not left until the night before submission.

Stage	What the team does	A college-portal example
Analysis	Write down who the users are and what the system must do.	Students must see the date sheet. Clerks must upload it.
Design	Decide screens, data, and structure before heavy coding.	A timetable table, a login screen, and an upload screen.
Coding	Build the design in a programming language.	Write the pages and the database queries.
Testing	Compare behaviour with the requirements.	Try a wrong password. Try a missing file.
Deployment	Put the system where real users reach it.	Publish it on the college server.
Maintenance	Fix faults and adapt to change.	Next year's class names are added.

Table 3. Stages you should be able to name and illustrate.

Two process models organise those stages differently. **Waterfall** completes each stage before the next begins. It is understandable and easy to document, and it fits when the requirements are stable, such as a payroll rule that is fixed by policy. It is painful when the first analysis was wrong, because the team discovers that late. **Agile** builds in short cycles called sprints. Each sprint delivers a small working piece, the user comments, and the next sprint adjusts. Agile fits a college app whose teachers keep changing what they want. It needs a customer who will actually look at each piece. Agile is not an excuse to skip testing.

You may see two further names on a key-term list. **UML** is a family of diagrams for design, such as a picture of users and their actions. A **design pattern** is a known shape for a recurring design problem. You are not required to draw a full UML model in this unit; you should know why a

diagram exists before the code.

Black-box and white-box testing

Black-box testing looks only at inputs and outputs. The tester does not read the code. If the requirement says “a mark below 0 is rejected”, the tester enters 1 and checks the message.

White-box testing looks inside. The tester chooses inputs that travel through each branch, including the branch that almost never runs. Black-box testing asks “did we build the right behaviour?” White-box testing asks “did we exercise the code we wrote?” A serious project uses both. Testing reduces risk. It does not prove that no bug remains.

EXAM TIP

A five-mark SDLC question wants the stage names in order, one activity for each, and a single running example. Do not define Agile in a question that asked for the stages, and do not list the stages in a question that asked for the difference between Waterfall and Agile.

Communication model, OSI, and TCP/IP

A basic **communication model** has a sender, an encoder that turns the message into a signal, a medium, a decoder, and a receiver. **Noise** is anything that damages the signal on the way: electrical interference, a damaged cable, or too many devices talking at once. A **protocol** is a shared rulebook: the order of messages, the addresses, and what to do when a piece is lost. Without a protocol, the bits arrive and neither side knows what they mean.

The **OSI model** of ISO splits the work into seven layers so that a change in the cable does not force a change in the email program. Learn the order from the wire upward, and one job plus one example for each layer.

Layer	Job	Examples
7 Application	The service the user actually wants.	HTTP, SMTP, FTP, DNS
6 Presentation	Format, compression, and encryption of the data.	Character encoding, TLS formatting
5 Session	Open, manage, and close a dialogue.	Keeping a login session in order
4 Transport	End-to-end delivery and ports.	TCP, UDP
3 Network	Addresses and routes across networks.	IP addresses, routers
2 Data link	Frames on one link, and MAC addresses.	Ethernet, a switch
1 Physical	The bits on the medium.	Cable, radio, hub, voltages

Table 4. OSI layers. PDU means the name of the chunk at that layer.

The **TCP/IP model** used on the real Internet folds those seven layers into four: Link (OSI 1–2), Internet (OSI 3), Transport (OSI 4), and Application (OSI 5–7). **IP** moves packets using IP addresses. **TCP** adds reliability: numbering, acknowledgements, and retransmission. **UDP** sends datagrams without that handshake, which is why it is used when a late packet is useless, such as a live voice call. A **port** number tells the receiving computer which program should get the segment.

COMMON MISTAKE

Switches and MAC addresses belong with the data-link layer. Routers and IP addresses belong with the network layer. HTTP is an application protocol that usually travels over TCP. Do not call HTTP a physical-layer protocol because you use it on a cable.

Topologies, scalability, and reliability

A **network topology** is the shape of the connections. The shape decides how easily the network grows (**scalability**) and how well it survives a fault (**reliability**). **Availability** is the share of time the service is actually usable: uptime divided by uptime plus downtime. A design can be fast and still unreliable if one small part stops everything.

Topology	Shape	Strength	Weakness
Bus	One shared backbone	Little cable	A break in the backbone affects every device; collisions grow as devices are added
Star	Each device to a central switch	Easy to add a device; one cable fault is local	The central switch is a single point of failure
Ring	Each device to the next, in a loop	Orderly access, often by a token	One break can stop the ring unless a second ring exists
Mesh	Many paths between devices	A failed link can be routed around	Many links, so high cost
Tree	Stars joined in a hierarchy	Matches a campus or an organisation	A failed link high in the tree isolates a whole branch
Hybrid	A mix of the above	Each part uses a suitable shape	Harder to document and to repair

Table 5. Topologies to compare in a long question.

On a bus, devices often share the cable using **CSMA**: listen before sending, and cope with a collision if two speak together. On a classic ring, a **token** is a permission that travels around the ring; only the device holding the token may send. In a **star**, a **client** uses a service and a **server** provides it. Those words also describe roles, not only topology: a lab PC can be a client of the college file server.

Cloud computing helps scale because more storage or processing can be rented when the load grows, and released when it falls. Reliability still has to be designed: two copies in two places, a spare path, and a test that actually adds load and removes a part to see what breaks. A test of scale is not the same thing as an attack. It is a measurement the owner runs on a system they are allowed to test.

Cybersecurity and encryption

A system is attacked because the data or the service is valuable. **Cybersecurity** is the work of keeping three properties: **confidentiality** (the wrong people cannot read it), **integrity** (nobody changes it unnoticed), and **availability** (the people who should reach it still can). A security plan ranks the risks and spends effort on the serious ones first. A written **privacy policy** says what data is collected and why. A policy does not protect anyone if the system ignores it.

Name	Impact	Defence to remember
Malware	Software that harms the device or the data	Install software from known sources; keep updates on
Phishing	A fake message that tries to collect a password or a payment	Check the real address; do not use a link from an unexpected message
DDoS	A service is flooded until genuine users cannot use it	The service owner filters traffic and adds capacity; users cannot stop this from a browser
XSS	A website runs a script a visitor did not intend, which can take over that visitor's session	The site must treat typed-in text as text, not as code
Unauthorised access	Someone uses an account or a system they do not own	Authentication, authorisation, and least privilege

Table 6. Threats named in the curriculum, described by impact. The defence column is what a student and a college should do.

Authentication checks who you are. **Authorisation** checks what that identity is allowed to do. A login proves identity; a clerk's account that cannot edit the date sheet is authorisation. **2FA** (two-factor authentication) asks for a second proof, such as a code on a phone, so a stolen password is not enough. A **firewall** allows or blocks traffic by rule. **Hashing** turns data into a short fingerprint. A good hash is one-way: you cannot rebuild the password from the fingerprint. That is why systems store hashes of passwords rather than the passwords themselves.

Encryption turns **plaintext** into **ciphertext** with a key, so that someone who sees the message still cannot read it. **Decryption** reverses the step with the proper key. **Symmetric** encryption uses the same secret key to lock and unlock. It is fast, and the hard part is delivering that secret key safely. **Asymmetric** encryption uses a key pair: a public key anyone may know, and a private key that stays secret. It is slower and is often used to agree a symmetric key, or to sign a message so the receiver can check who sent it. Hashing is not encryption. A hash is not meant to be decrypted.

WORKED EXAMPLE — A TEACHING CIPHER, NOT A REAL DEFENCE

The Caesar cipher shifts each letter by a fixed number. Shift BOARD by 3: BE, OR, AD, RU, DG, so the ciphertext is ERDUG. Shifting back by 3 restores BOARD. This shows plaintext, ciphertext, and a key (the number 3). It is a classroom illustration only. A shift cipher is easy to undo by trying all 25 shifts, so it does not protect a password, a result, or a message.

WHAT THIS LECTURE WILL NOT TEACH YOU TO DO

You should be able to name an attack, say what harm it causes, and say which control reduces that harm. Building a phishing page, flooding a network, or writing an exploit is not part of the practical and is not a skill this course awards marks for.

Key terms

Term	Meaning in this lecture
Bit / byte	A bit is a 0 or a 1. A byte is eight bits.
Boolean function	A rule that gives one output bit for each combination of input bits.
Duality	Swap AND with OR and 0 with 1 in a law. The dual law is also true.
Logic gate	A circuit element that performs AND, OR, NOT, NAND, NOR, XOR, or XNOR.
Karnaugh map	A truth table arranged so groups of 1s can be simplified.
SDLC	The staged process of analysis, design, coding, testing, deployment, and maintenance.
Waterfall / Agile	One-pass stages, versus short cycles with user feedback.
Black-box / white-box	Testing from the outside by inputs and outputs, versus testing internal paths.
Protocol	An agreed set of rules for communication.
OSI model	Seven layers from physical up to application.
TCP/IP	The four-layer model of the Internet: link, internet, transport, application.
Topology	The shape of a network: bus, star, ring, mesh, tree, or hybrid.
Scalability	The ability to grow in users or data without collapsing.
Reliability	The ability to keep working when a part fails.
Encryption	Reversible transformation of plaintext into ciphertext using a key.
Hashing	A one-way fingerprint, used to detect change and to store password checks.
Authentication	Checking identity. Authorisation then checks permission.

Check your understanding

Attempt each question before you read the model answer. In the examination, a short question usually earns marks for a definition plus one precise point or a tiny example.

Why is a digital copy cleaner than a long chain of analog copies?

MODEL ANSWER

Each digital stage restores the signal to a definite 0 or 1, so small noise is thrown away. An analog copy copies the noise as well, so quality falls at every step.

State the dual of $A + 1 = 1$.

MODEL ANSWER

Swap $+$ with $\cdot$ and 1 with 0. The dual is $A \cdot 0 = 0$.

A gate outputs 0 only when both inputs are 1, and outputs 1 otherwise. Name it.

MODEL ANSWER

NAND. It is the NOT of AND, and AND is 1 only when both inputs are 1.

Give one activity of the design stage and one activity of the testing stage.

MODEL ANSWER

Design decides the structure, for example the tables and the screens. Testing checks the running system against the requirements, for example by entering a bad password and reading the result.

Which OSI layer uses IP addresses, and which uses MAC addresses?

MODEL ANSWER

The network layer uses IP addresses and routers. The data-link layer uses MAC addresses and switches.

Board-style practice

Multiple choice

Each item is worth one mark. Cross out two options that cannot be right before you choose.

1. Which identity is the dual of $A \cdot 1 = A$?

- A. $A + 0 = A$
- B. $A + 1 = 1$
- C. $A \cdot 0 = 0$
- D. $A \cdot A = A$

2. The output of XOR is 1 when

- A. both inputs are 1
- B. both inputs are 0
- C. the inputs are different
- D. either input is 1, including both

3. Black-box testing mainly checks

- A. the names of the variables
- B. inputs against expected outputs
- C. the brand of the keyboard
- D. how many comments are in the code

4. In the TCP/IP model, TCP belongs to the

- A. link layer
- B. internet layer

- C. transport layer
- D. physical cable

5. A star topology is easy to extend, but

- A. it has no central device
- B. the central switch is a single point of failure
- C. it cannot use a switch
- D. it is always more reliable than a mesh

6. Hashing a password is preferred to storing the password because

- A. a hash is designed to be reversed with the public key
- B. a hash is a one-way fingerprint
- C. a hash makes the network faster
- D. a hash replaces the need for authentication

Answer key with reasons

Q	Answer	Why
1	A	Duality swaps · with + and 1 with 0, so $A \cdot 1 = A$ becomes $A + 0 = A$.
2	C	XOR is 1 only on rows 01 and 10. Row 11 is 0, so option D is ordinary OR.
3	B	The tester does not need the source. The requirement and the observed output are enough.
4	C	TCP provides end-to-end reliable delivery, which is the transport job.
5	B	Every device depends on the centre. A mesh has extra paths and is usually more reliable, at a higher cost.
6	B	The system can check a login by hashing the typed password and comparing fingerprints. It need not keep the password itself.

Short questions

S1. Differentiate analog and digital signals with one example of each.

MODEL ANSWER

An analog signal varies smoothly, such as the voltage from a microphone. A digital signal uses discrete values, such as the bits stored in a computer. A digital system samples and quantises an analog source before it can store it.

S2. Draw the truth table of NAND and state why NAND is called universal. (A description is enough if the paper does not provide a drawing box.)

MODEL ANSWER

Rows 00, 01, 10, 11 give outputs 1, 1, 1, 0. NAND is universal because NOT, AND, and OR can all be constructed from NAND gates alone, so any Boolean function can be built from NAND.

S3. Give two differences between Waterfall and Agile.

MODEL ANSWER

Waterfall finishes a stage before the next starts; Agile repeats short cycles. Waterfall suits fixed requirements; Agile expects the requirements to be refined after each working increment. Both still need testing.

S4. What is the difference between symmetric and asymmetric encryption?

MODEL ANSWER

Symmetric encryption uses one shared secret key to encrypt and decrypt, and it is fast, but the key must be delivered safely. Asymmetric encryption uses a public key and a private key. It is slower and is used to exchange a secret or to check a signature.

Long questions

L1. (5 marks) A college wants a result portal. Relate the SDLC stages to this project, and say whether you would start with Waterfall or Agile. Justify the choice.

MODEL ANSWER

Analysis lists the users and the rules: students view their own result, teachers enter marks, and the controller publishes them. Design chooses the data (student, subject, mark) and the screens. Coding builds those screens and the checks, such as rejecting a mark above 100. Testing tries correct marks, out-of-range marks, and a student opening another student's roll number. Deployment places the portal on the college server. Maintenance covers a changed grade scale next year.

Agile is the better start if teachers are still deciding the screens, because a first sprint can publish a read-only result and the next sprint can add mark entry. Waterfall would fit better only if the board's rules and the screens were already frozen. Name the risk either way: a wrong roll-number check is a privacy fault, so that test belongs in the plan before launch.

L2. (5 marks) Compare star and mesh topologies for a computer laboratory of 40 PCs that must stay usable during a practical examination.

MODEL ANSWER

A star joins each PC to one switch. Adding a PC is one cable, a broken PC cable does not stop the others, and fault finding is simple. The weakness is the switch: if it fails, the whole lab stops, which is unacceptable in an examination. A full mesh would give each PC many paths, so one failed link would not isolate the lab, but 40 PCs cannot sensibly be fully meshed; the cable count is enormous.

A practical design is a star, or a tree of stars, with a second switch ready, or two switches so that the server remains reachable. Reliability here means an examination can continue after one failure. Scalability means next year's larger class can be added without replacing the whole lab. The pure mesh wins on paper for reliability and loses on cost. The examined answer should say that trade-off in one sentence.

Practical work

The practical paper is 25 marks. Keep a journal: aim, steps, the program or diagram, the output, and one sentence on what went wrong the first time.

Lab 1 — Truth table and logic diagram

- 1 On paper, draw $F = (A \text{ AND } B) \text{ OR } (\text{NOT } C)$. Label every wire.
- 2 Complete the eight-row truth table in binary order.
- 3 If Logisim (or Logisim-evolution) is installed, build the same circuit, toggle the inputs, and tick each row your table got right.
- 4 Change the expression to $F = (A \text{ NAND } B)$. Record the new table and name the gate.

WHAT SUCCESS LOOKS LIKE

Your paper table matches the simulator on all eight rows, or, if you have no simulator, a classmate can recompute two rows you chose at random and get your answers.

Lab 2 — SDLC on one page

- 1 Pick a small system used in college: attendance, library issue, or the result portal.
- 2 Write six headings for the SDLC and two bullets under each, specific to that system.
- 3 Write three black-box tests and one white-box idea (a branch that must be tried, such as an empty name).
- 4 State one risk and how the test plan catches it.

WHAT SUCCESS LOOKS LIKE

A reader who was not in the room can tell what will be built and how you will know it failed.

Copy this summary

- Digital signals use discrete bits; analog signals vary smoothly and pick up noise when copied.
- Learn the identities, De Morgan's laws, and duality. Duality swaps operators and constants; it does not claim the dual expression has the same value.
- NAND outputs 0 only when all inputs are 1. XOR outputs 1 when the inputs differ.
- Three-input tables have eight rows. Work in columns.
- SDLC: analyse, design, code, test, deploy, maintain. Waterfall is one pass. Agile is short cycles. Test from the outside and from the inside.
- OSI has seven layers. TCP/IP has four. IP addresses routes; TCP adds reliability; a switch is not a router.
- Star is easy and centrally fragile. Mesh is resilient and costly. Scalability and reliability are different compliments.
- Encryption is reversible with a key. Hashing is a one-way fingerprint. Authentication is not the same word as authorisation.

CLASS XI · UNIT 2

Lecture 2

Computational Thinking and Algorithms

Textbook unit. Unit 2 — Computational Thinking and Algorithms

Curriculum domain. B. Computational thinking and algorithms

Class time. about 20 periods

Computational thinking is the habit of turning a messy problem into steps a computer can follow. This lecture covers the four habits — decomposition, pattern recognition, abstraction, and algorithms — then the two families of algorithms the paper actually asks you to trace: searching and sorting.

Learning outcomes

- Analyse a problem using decomposition, pattern recognition, and abstraction.
- State the properties of a proper algorithm and write clear pseudocode.
- Trace a flowchart or a pseudocode listing and judge correctness, clarity, and efficiency in plain language.
- Apply linear search, binary search, bubble sort, and insertion sort to a small list and show the steps.

Four habits of computational thinking

Decomposition breaks one problem into smaller problems that can be solved separately. A result portal decomposes into login, mark entry, calculation, and display. **Pattern recognition** notices that two of those pieces are the same kind of check: both reject an empty box, both reject a value out of range. **Abstraction** keeps the detail that matters and hides the rest. For a search, the list items matter and the colour of the screen does not. An **algorithm** is the finite sequence of precise steps that uses those habits.

WORKED EXAMPLE — COUNTING A REAL PROBLEM

Problem: how many students in this class scored 80 or above?

Decomposition: get the list, look at one mark, decide, count, move on, stop at the end. Pattern: every mark is handled the same way, so a loop is honest. Abstraction: the algorithm needs the marks, not the students' addresses. A vague instruction such as “look at the good ones” is not an algorithm, because two people would not do the same thing.

What makes an algorithm acceptable

A classroom algorithm should have five properties. It has **input**. It has **output**. Each step is **definite** (no guesswork). It is **finite** (it ends). Each step is **effective** (a person, or a computer, can actually carry it out). “Try every possible password in the world until something happens” fails finiteness in any useful sense and is not a task this course sets.

Pseudocode is structured English, not a programming language. For board work, use line numbers, a consistent indent for the body of a loop or a decision, uppercase words for the commands, and a short comment only where the step is not obvious. Declare the data in one line before you use it. The usual commands in this course are INPUT, SET, IF, ELSE, WHILE, FOR, and OUTPUT.

Pseudocode

```

1 INPUT mark
2 IF mark >= 80 THEN
3     OUTPUT "Distinction"
4 ELSE
5     IF mark >= 40 THEN
6         OUTPUT "Pass"
7     ELSE
8         OUTPUT "Needs improvement"
9     ENDIF
10 ENDIF

```

A **flowchart** is the same algorithm in symbols. An oval starts and stops. A parallelogram is input or output. A rectangle is a process. A diamond is a decision, and every diamond needs two labelled exits. Arrows must not vanish. Tracing means writing the value of each variable after every step, on a fresh row, until you reach the end. A trace that skips the false branch has not tested the algorithm.

Correctness, clarity, and efficiency

An algorithm is **correct** if, for every allowed input, the output matches the specification. One lucky example does not prove correctness; a single counter-example disproves it. It is **clear** if another student can trace it without asking you what a line means. Names such as count and position are clearer than x and y once a problem has two different numbers. It is **efficient**, at this level, if it does not repeat work for no reason. You do not need formal big-O notation to say that looking at every item once is kinder than looking at every pair of items, when a single pass is enough.

COMMON MISTAKE

A flowchart diamond with only one arrow is unfinished. An IF in pseudocode without an ENDIF, or an indent that does not match the ENDIF, will be marked as unclear even when your idea was right.

Linear search

Linear search starts at the first item and compares until it finds the target or runs out of items. The list need not be sorted. If the target is absent, the honest result is “not found”, not the last item.

Pseudocode

```

1 INPUT list, n, target
2 SET position TO 0
3 SET index TO 1
4 WHILE index <= n AND position = 0
5     IF list[index] = target THEN
6         SET position TO index
7     ELSE
8         SET index TO index + 1
9     ENDIF
10 ENDWHILE
11 IF position = 0 THEN
12     OUTPUT "Not found"
13 ELSE
14     OUTPUT position
15 ENDIF

```

Positions in that listing start at 1, which is the usual board convention. Python later in the course starts at 0. Say which convention you are using. Trace of target 4 in the list 5, 1, 4, 2: index 1 compares 5, index 2 compares 1, index 3 compares 4 and sets position to 3. Three comparisons. If the target were 9, the loop would examine all four items and output “Not found”.

Binary search

Binary search requires a **sorted** list. It compares the target with the middle item. If the target is smaller, the right half is discarded. If it is larger, the left half is discarded. Each comparison removes about half of the remaining items, which is why it beats linear search on a long sorted list. On an unsorted list it is simply wrong, not “a bit slower”.

WORKED EXAMPLE — BINARY SEARCH FOR 16

Sorted list, positions 1 to 7: 2, 5, 8, 12, 16, 23, 38. Target 16.

Low = 1, high = 7. Middle position = $(1+7) \div 2 = 4$, value 12. 16 is greater, so low becomes 5.

Low = 5, high = 7. Middle = $(5+7) \div 2 = 6$, value 23. 16 is smaller, so high becomes 5.

Low = 5, high = 5. Middle = 5, value 16. Found at position 5.

Use integer division, discarding the fraction. Write low, high, and middle on every row of the trace.

If the item is absent, low eventually passes high and you stop. Searching for 10 in the same list ends with the middle values 12, then 8, then 12's neighbour is exhausted, and the result is not found. Show that stopping condition in a trace if the question asks for a missing item.

Bubble sort

Bubble sort repeatedly walks the list and swaps any neighbouring pair that is out of order. After the first pass the largest item has bubbled to the end. After the second pass the next largest is in place. You may stop early if a pass makes no swaps.

WORKED EXAMPLE — BUBBLE SORT OF 5, 1, 4, 2

Pass 1: compare 5 and 1, swap to 1, 5, 4, 2. Compare 5 and 4, swap to 1, 4, 5, 2. Compare 5 and 2, swap to 1, 4, 2, 5.

Pass 2: compare 1 and 4, keep. Compare 4 and 2, swap to 1, 2, 4, 5. Compare 4 and 5, keep.

Pass 3: 1 and 2 are in order, 2 and 4 are in order. No swap, so the list is sorted: 1, 2, 4, 5.

The end of each pass is the line the examiner wants to see. Do not jump to the final list.

Insertion sort

Insertion sort keeps a sorted portion on the left and inserts the next item into its proper place, shifting larger items right to make a gap. It is the way many people sort a hand of cards.

WORKED EXAMPLE — INSERTION SORT OF 5, 1, 4, 2

Start with the first card: [5] | 1, 4, 2.

Insert 1 in front of 5: [1, 5] | 4, 2.

Insert 4 between 1 and 5: [1, 4, 5] | 2.

Insert 2 between 1 and 4: [1, 2, 4, 5].

Each line shows the sorted portion in brackets. That is the trace.

Choose the algorithm to match the data. An unsorted list of 12 names on a worksheet is a linear search or an insertion sort; binary search would be a mistake until you sort. A sorted attendance roll of several hundred names is a binary search. Bubble sort is easy to trace in an examination and wasteful on a huge list, which is a fair comment if a question asks you to evaluate it. Clarity is why we still teach the trace.

EXAM TIP

If the question says “show each pass”, a final sorted list with no passes scores almost nothing. If it says “which algorithm”, the first sentence must name it and the second must give the reason, usually “the list is not sorted” or “the list is already sorted and long”.

Key terms

Term	Meaning in this lecture
Decomposition	Splitting a problem into smaller problems.
Pattern recognition	Noticing that the same step repeats or that two problems share a shape.
Abstraction	Keeping the necessary detail and ignoring the rest.
Algorithm	A finite, precise sequence of steps from input to output.
Pseudocode	Structured English for an algorithm, with indent and line numbers.
Trace	A table of variable values after each step.
Linear search	Check items one by one. The list may be unsorted.
Binary search	Halve a sorted list at each comparison.
Bubble sort	Swap neighbouring items that are out of order, pass after pass.
Insertion sort	Insert each next item into an already sorted left-hand portion.

Check your understanding

Attempt each question before you read the model answer. In the examination, a short question usually earns marks for a definition plus one precise point or a tiny example.

Why is “sort the names nicely” not an algorithm?

MODEL ANSWER

It is not definite. Two people can interpret “nicely” differently, so the steps are not precise and the result is not determined.

Can binary search be used on 9, 1, 4, 7 to find 4?

MODEL ANSWER

No. The list is not sorted. Sort it first, or use linear search. Binary search on unsorted data can report “not found” when the item is present.

After the first full pass of bubble sort on 5, 1, 4, 2, what is the list?

MODEL ANSWER

1, 4, 2, 5. The largest value, 5, has moved to the end.

Board-style practice

Multiple choice

Each item is worth one mark. Cross out two options that cannot be right before you choose.

1. Which property fails if an algorithm can run forever on a valid input?

- A. Input
- B. Output
- C. Finiteness
- D. Abstraction

2. Binary search is appropriate when the data is

- A. unsorted and small
 - B. sorted
 - C. a single number
 - D. stored only on paper and must not be ordered
3. In insertion sort, the left-hand portion after each insertion is
- A. the unsorted remainder
 - B. sorted
 - C. reversed
 - D. always a single item
4. Abstraction, in the counting-marks problem, means
- A. collecting home addresses as well as marks
 - B. ignoring details that the steps do not use
 - C. writing the program in Python immediately
 - D. sorting before you count

Answer key with reasons

Q	Answer	Why
1	C	Finiteness means the steps end. A loop with no progress toward a stopping condition fails it.
2	B	Each step throws away half of the list, which is valid only if every item on one side is smaller and every item on the other side is larger.
3	B	That is the idea of the algorithm: grow a sorted portion one item at a time.
4	B	The count depends on the marks. Addresses are irrelevant detail.

Short questions

S1. List the four computational-thinking habits and one sentence for each.

MODEL ANSWER

Decomposition splits the problem. Pattern recognition finds repeated structure. Abstraction drops detail that does not affect the result. An algorithm writes the remaining work as finite precise steps.

S2. Trace linear search for 8 in the list 3, 8, 8, 1. Positions start at 1. Which 8 do you report, and why?

MODEL ANSWER

Compare position 1 (3), then position 2 (8). Stop and report position 2. The second 8 is never examined. The algorithm as written returns the first match, so say so.

S3. Give one reason bubble sort is taught, and one reason it is a poor choice for a very large list.

MODEL ANSWER

It is taught because each pass is easy to trace by hand and the idea of a swap is visible. It is a poor choice at large scale because it repeatedly compares neighbours and moves values only one place at a time, so the number of comparisons grows much faster than the length of the list.

Long questions

L1. (5 marks) The sorted list is 4, 9, 15, 21, 28, 33, 40. Trace binary search for 21. Then explain, without a formula, why this method beats linear search as the list grows, and when it must not be used.

MODEL ANSWER

Positions 1 to 7. Low 1, high 7, middle 4, value 21. The middle value is the target, so the search stops at position 4. If a marker scheme expects you to show the halving anyway, state that the first comparison already succeeded, so there is no second row. Do not invent extra steps.

Linear search may have to look at every item. Binary search throws away half of the remaining items after each comparison, so a list that is twice as long needs only one extra comparison, not twice the work. It must not be used when the list is unsorted, because the decision to discard a half depends on every item on that side being too small or too large.

L2. (5 marks) Sort 7, 3, 7, 1 by insertion sort, showing the sorted portion after each insertion. Then say how your trace would differ if the question had asked for bubble sort.

MODEL ANSWER

Insertion: [7] | 3, 7, 1. Insert 3 to get [3, 7] | 7, 1. Insert 7, which stays after the existing 7, to get [3, 7, 7] | 1. Insert 1 at the front to get [1, 3, 7, 7]. Equal values keep their relative order in this left-to-right insertion.

Bubble sort would instead show full passes of neighbour swaps. Pass 1 ends as 3, 7, 1, 7. Pass 2 ends as 3, 1, 7, 7. Pass 3 ends as 1, 3, 7, 7. The final list matches, but the intermediate lines do not, so a bubble-sort trace written as insertion steps would lose the method marks.

Practical work

The practical paper is 25 marks. Keep a journal: aim, steps, the program or diagram, the output, and one sentence on what went wrong the first time.

Lab 3 — Trace before you code

- 1 Write pseudocode for binary search with positions starting at 1.
- 2 Trace it for the list 2, 5, 8, 12, 16, 23, 38 and the target 23. Keep a three-column table: low, high, middle value.
- 3 Trace the same algorithm for the target 7 and show the step where low passes high.
- 4 Ask a partner to follow only your pseudocode, not your explanation. Correct any line they cannot execute.

WHAT SUCCESS LOOKS LIKE

The partner's trace matches yours, including the not-found stop, without extra oral help.

Copy this summary

- Decomposition, patterns, and abstraction come before code.
- An algorithm has input, output, definite steps, an end, and steps that can be carried out.
- Pseudocode needs indent, end markers, and line numbers. A decision symbol needs two exits.
- One counter-example destroys a claim of correctness. Clarity is part of the mark.
- Linear search walks from the start and works on unsorted data. It returns the first match.
- Binary search halves a sorted list. Integer middle, and stop when low passes high.
- Bubble sort swaps neighbours. Show every pass. Insertion sort grows a sorted left portion.

CLASS XI · UNIT 3

Lecture 3

Programming Fundamentals in Python

Textbook unit. Unit 3 — Programming Fundamentals

Curriculum domain. C. Programming fundamentals

Class time. about 40 periods

A program takes input, processes it, and produces output, using the same processor and memory you described in computer systems. Python is the language of this unit because the programs are short, the errors are readable, and the same language continues in Class XII. You will write sequence, selection, and repetition; draw with the turtle; use a library without reading its source; store values in a list; and split a large job into functions you can test.

Learning outcomes

- Explain what a program does with input, process, and output, and name the parts of an IDE.
- Write and run Python that uses variables, operators, decisions, and loops.
- Draw compound shapes with the turtle, including a pen-up move.
- Use a standard library and explain abstraction as the reason you can.
- Populate a list, walk through it, and find an item.
- Decompose a problem into functions with parameters and return values.
- Find a fault by a dry run and by a breakpoint or a printed check.

Why programs exist

Hardware does nothing useful until software tells it which input to read, which calculation to repeat, and which output to show. Python is used for science, small web services, data work, and teaching, because the same ideas — variables, decisions, loops, functions — appear in larger systems. A real task for this year is modest: a fee receipt, a pass-fail list, a square drawn on screen. The habit that matters is that the program matches a specification you could have written in pseudocode first.

An **IDE** (integrated development environment) is the workbench. You need an editor, a way to run the file, a console for input and output, a file list, and a debugger. Thonny, IDLE, and VS Code are offline. A site such as Replit is an online IDE when the lab has no install rights. The language is the same. Learn where the Run button is and where error messages appear: they name a line number, and that line is the first place you look, not the last.

Values, variables, and operators

A **variable** is a name bound to a value. `marks = 78` stores an integer. `fee = 2500.50` stores a float. `grade = "B"` stores a string. `passed = True` stores a Boolean. Python decides the type from the value. `input()` always returns a string, so a numeric answer must be converted with `int()` or `float()` before you do arithmetic.

Python

```
name = input("Name: ")
obtained = int(input("Marks obtained: "))
total = int(input("Total marks: "))
percent = obtained / total * 100
print(name, round(percent, 1))
```

Arithmetic operators are `+` `-` `*` `/` `//` `%` `**`. `//` is integer division. `%` is the remainder. `**` is power. Comparison operators are `==` `!=` `<` `>` `<=` `>=`. They produce True or False. Logical operators are `and`, `or`, and `not`. Assignment is `=`. Comparison is `==`. Mixing them up is the most common first-month bug.

COMMON MISTAKE

`if marks = 40` is a syntax error because `=` assigns. The test is `if marks == 40`. And `int("78.5")` fails: a decimal string needs `float`, not `int`.

Sequence, selection, and repetition

Sequence means the lines run from top to bottom. **Selection** chooses a branch with `if`, `elif`, and `else`. **Repetition** repeats a block with `while` or `for`. The body of each of these is the indented lines underneath. Four spaces is the usual indent. A missing indent is a syntax error. An extra indent attaches a line to the wrong block, which is worse, because the program runs and does the wrong thing.

Python

```
marks = int(input("Marks: "))
if marks >= 80:
    result = "Distinction"
elif marks >= 40:
    result = "Pass"
else:
    result = "Needs improvement"
print(result)
```

A `while` loop needs a condition that eventually becomes false. A `for` loop is the better tool when you know how many times to repeat, or when you are walking through a list. `range(1, 6)` produces 1, 2, 3, 4, 5. The stop value is not included.

Python

```
total = 0
for subject in range(1, 6):
    score = int(input("Subject " + str(subject) + ": "))
    total = total + score
print("Total", total, "Average", total / 5)
```

Turtle graphics

The **turtle** is a pen that moves on a screen. You give it distances and angles. `forward` and `backward` move. `right` and `left` turn, in degrees. `penup` lifts the pen so a move does not draw. `pendown` puts it back. `circle` draws a circle. Colour and width are properties of the pen. A compound shape is several simple shapes, with the pen lifted between them so you do not draw a stray line.

Python (turtle)

```
import turtle

pen = turtle.Turtle()
pen.color("navy")
for _ in range(4):
    pen.forward(120)
    pen.right(90)
pen.penup()
pen.goto(30, 30)
pen.pendown()
pen.color("teal")
pen.circle(30)
turtle.done()
```

The square uses repetition: four sides, each followed by a right angle. The circle is a second shape. `penup` before `goto` is what keeps the two shapes separate. `turtle.done()` keeps the window open until the user closes it. In a written paper, you may be asked to state the angle inside the loop. A regular pentagon turns $360 / 5 = 72$ degrees, not 90.

Libraries and abstraction

A **library** is a collection of functions somebody else has tested. `import math` gives you `math.sqrt` and `math.pi`. You use them without reading how the square root is calculated. That is **abstraction** in programming: a name hides a complicated implementation behind a small, stable interface. A third-party library is installed separately and then imported the same way. You choose one because it solves a real sub-problem, not because it is famous.

Python

```
import math

radius = float(input("Radius: "))
area = math.pi * radius ** 2
print(round(area, 2))
print(math.sqrt(area))
```

BUILT-IN AND THIRD-PARTY

`math`, `random`, and `turtle` ship with Python. A library such as a charting package may need an install. In the examination you can assume `math` and `turtle` exist. Say `import` in the first lines of any program that uses them.

Lists

A **list** stores an ordered series of values. Indexes start at 0. `marks[0]` is the first mark. `append` adds at the end. `len` gives the length. A loop can populate a list, and a second loop can search it. This is linear search from the previous lecture, now in Python.

Python

```
marks = []
for _ in range(4):
    marks.append(int(input("Mark: ")))

target = int(input("Find: "))
found_at = -1
for index in range(len(marks)):
    if marks[index] == target and found_at == -1:
        found_at = index

if found_at == -1:
    print("Not found")
else:
    print("Index", found_at)
```

If the user enters 10, 20, 20, 30 and searches for 20, the index printed is 1, the first match. Index 1 is a sentinel meaning “not found”. Do not use 0 as the sentinel: 0 is a legal index.

Functions

A **function** is a named block with a job. You **define** it with `def` and you **call** it by name. **Parameters** are the names in the definition. **Arguments** are the values in the call. A **return** sends a value back. Code that is copied in three places should be moved into one function so a fix happens once. This is decomposition: the large problem calls smaller problems.

A name created inside a function is **local**. It is not visible outside. A parameter is also local. This course passes numbers and strings **by value** in the everyday sense that changing the parameter does not change the caller’s variable, unless you return the new value and the caller stores it. (If you later pass a list and append to it, the list is shared. For the functions in this unit, return the result.)

Python

```
def celsius_to_fahrenheit(celsius):
    return celsius * 9 / 5 + 32

def fee_total(items, rate):
    return items * rate

print(celsius_to_fahrenheit(0))
print(celsius_to_fahrenheit(100))
print(fee_total(3, 250))
```

The calls print 32.0, 212.0, and 750. A function that prints but does not return is awkward to test, because the result is stuck on the screen. Prefer returning a value, and let the caller print it. Functions that call other functions, or that are called inside a loop, are normal: a loop of five temperatures can call the converter five times.

Debugging

A **syntax error** stops the program before it runs: a missing colon, a bad indent, a bracket left open. A **logic error** runs and gives the wrong answer. The second kind is why we test. Code inside a function is easier to test because you can call it with a known input and look at the returned value. A long script that only runs from `input()` makes you type the same data on every attempt.

A **dry run** is a paper trace of the variables. A **breakpoint** is a mark in the debugger where execution pauses so you can see those variables in the middle of a run. Until you are fluent with the debugger, a temporary `print` of the index and the value is an honest tool. Remove those prints before you submit the practical, or label them clearly as checks.

WORKED EXAMPLE — A FAULT IN A LOOP

A student writes `for index in range(1, len(marks))` and never sees the first mark. `range` starts at 1, but the first index is 0, so `marks[0]` is skipped. The dry run of the list `[40, 50]` shows index 1 only. The fix is `range(len(marks))`. The same trace would have caught an off-by-one in pseudocode last lecture.

EXAM TIP

In a program question, write the imports, then the functions, then the calls. Show one sample run underneath: the input you imagined and the output the code produces. Markers give method marks for a sensible structure even if a colon is missing, and output marks only when the sample is consistent with the code.

Key terms

Term	Meaning in this lecture
IDE	Editor, runner, console, file list, and debugger in one place.
Variable	A name bound to a value.
Selection	<code>if</code> , <code>elif</code> , and <code>else</code> .
Repetition	<code>while</code> or <code>for</code> , with an indented body.
Turtle	A drawing pen controlled by distances, turns, and pen-up.
Library	Tested code you import and use through a small set of names.
List	An ordered collection. The first index is 0.
Function	A named, testable block. Parameters in, a return value out.
Scope	Where a name is visible. A variable inside a function is local.
Breakpoint	A pause in the debugger so you can inspect variables.

Check your understanding

Attempt each question before you read the model answer. In the examination, a short question usually earns marks for a definition plus one precise point or a tiny example.

What does `range(5)` produce?

MODEL ANSWER

0, 1, 2, 3, 4. Five numbers, starting at 0, stopping before 5.

Why must the result of `input()` be converted before you compare it with 40?

MODEL ANSWER

`input()` returns a string. Comparing the string "40" with the number 40 is not the numeric test you wanted, and subtracting from a string is an error.

What is the index of the value 8 in `[8, 3, 8]` if your loop returns the first match?

MODEL ANSWER

0. The later 8 is at index 2 and is not returned.

Board-style practice

Multiple choice

Each item is worth one mark. Cross out two options that cannot be right before you choose.

1. Which line correctly tests equality?

- A. `if marks = 40:`
- B. `if marks == 40:`
- C. `if marks eq 40:`
- D. `if marks := 40`

2. To move the turtle without drawing, you call

- A. `penup()`
- B. `hide()`
- C. `color("white")`
- D. `break`

3. A function returns a value with

- A. `print`
- B. `return`
- C. `input`
- D. `import`

4. Using `math.sqrt` without studying its source is an example of

- A. decomposition only
- B. abstraction
- C. a syntax error
- D. white-box testing

Answer key with reasons

Q	Answer	Why
1	B	<code>==</code> compares. A single <code>=</code> assigns and is illegal in this <code>if</code> .
2	A	<code>penup</code> lifts the pen. Changing colour still draws.
3	B	<code>print</code> shows text. <code>return</code> hands a value back to the caller.

Q	Answer	Why
4	B	The name hides the implementation behind an interface you can trust.

Short questions

S1. Name four parts of an IDE and the job of each.

MODEL ANSWER

The editor holds the code. The runner executes the file. The console shows output and accepts input. The debugger pauses at a breakpoint so variables can be inspected. A file list is the fifth, useful part.

S2. Write a function that takes a Celsius value and returns Fahrenheit. Show the call for 0.

MODEL ANSWER

`def celsius_to_fahrenheit(celsius): return celsius * 9 / 5 + 32.` The call `celsius_to_fahrenheit(0)` returns 32.0. The formula is the familiar multiply by 9/5 and add 32.

S3. Explain one logic error that a dry run catches and a syntax checker does not.

MODEL ANSWER

Starting a list loop at 1 instead of 0 is valid Python, so the program runs. The dry run shows that the first item is never examined. The syntax checker has nothing to report.

Long questions

L1. (5 marks) A stationer sells copies at a given rate and charges a flat packing fee. Write a Python program that defines a function for the bill, reads the number of copies, the rate, and the packing fee, and prints the total. Explain how you would test the function.

MODEL ANSWER

```
def bill(copies, rate, packing): return copies * rate + packing
copies = int(input("Copies: ")) rate = float(input("Rate: ")) packing = float(input("Packing: "))
print(bill(copies, rate, packing))
```

Test the function without input first. `bill(2, 100, 20)` must return 220. `bill(0, 100, 20)` must return 20. A wrong formula such as adding the packing inside the multiplication fails the second test. Because the function returns a number, the test does not depend on reading the screen.

L2. (5 marks) Describe a turtle program that draws a square of side 100 and then a circle that does not sit on the square's outline. State the role of the loop and of penup.

MODEL ANSWER

Import turtle and create a turtle. Loop four times: forward 100, right 90. That repetition is the square; the angle is 90 because a square turns a full 360 degrees in four equal turns. Then penup, goto a point inside or clearly beside the square, pendown, and circle with a radius small enough not to cross the outline, for example 20. penup is required so the move between the shapes does not leave a line. pendown is required before the circle or the circle will not appear. A compound drawing is just sequence plus one loop, with the pen state managed between shapes.

Practical work

The practical paper is 25 marks. Keep a journal: aim, steps, the program or diagram, the output, and one sentence on what went wrong the first time.

Lab 4 — Decisions, loops, and a function

- 1 Type the marks program with if, elif, and else. Run it for 80, 40, and 39.
- 2 Write `celsius_to_fahrenheit` and print the results for 0 and 100 before you add `input()`.
- 3 Build a list of five integers using a loop and `append`. Search for a value that is present and a value that is absent.

- 4 Dry-run the search on paper for the present value and compare it with the program's index.

WHAT SUCCESS LOOKS LIKE

0 °C prints 32.0 and 100 °C prints 212.0. The search index matches your paper trace, and the missing value prints Not found.

Lab 5 — Turtle compound shape

- 1 Draw a square using a loop.
- 2 Lift the pen, move, lower the pen, and draw a circle.
- 3 Change the square into a regular hexagon by turning 60 degrees. Record the angle you chose and why.

WHAT SUCCESS LOOKS LIKE

The two shapes are separate, the hexagon closes, and your journal states that the exterior turn is 360 divided by the number of sides.

Copy this summary

- A program is input, process, and output. The IDE is where you edit, run, and debug.
- `input()` returns text. Convert it before arithmetic. `==` compares; `=` assigns.
- Indentation is the structure. `elif` is the second test. `range` stops before the end value.
- Turtle: forward, turn, penup, pendown, circle. A regular polygon turns 360 divided by the number of sides.
- Importing `math` is abstraction: you use a name and not the internal method.
- List indexes start at 0. Do not use 0 as a not-found flag.
- Functions take parameters and return results. Test the return value. Dry-run logic errors; the syntax checker will not see them.

CLASS XI · UNIT 4

Lecture 4

Data and Analysis

Textbook unit. Unit 4 — Data and Analysis

Curriculum domain. D. Data and analysis

Class time. about 25 periods

This unit joins two ideas the textbook treats together: keeping data in a database so it stays consistent, and looking at data so a claim is more than a feeling. You will model a small college database, then compute a summary, draw an honest chart, and say the difference between a pattern and a cause.

Learning outcomes

- Explain tables, keys, relationships, referential integrity, and an entity-relationship model.
- Choose a sensible data type and describe forms, queries, and reports.
- Compute mean, median, and mode, and match a chart to a question.
- Describe a straight-line model $y = mx + c$ and separate correlation from causation.
- Distinguish population and sample, parameter and statistic, and an experiment from an observation.

From lists of files to a database

A **database** is an organised collection of related data. A **database management system** (DBMS), such as the relational tools used in the lab, stores that data, checks the rules, and answers questions. A folder of spreadsheets can hold the same facts, and then the same student is spelled three ways and nobody knows which phone number is current. The database's job is one copy of each fact, with rules that reject nonsense.

A **table** is about one kind of thing. A **row** (record) is one instance. A **column** (field) is one fact about it. A **primary key** identifies a row and does not repeat: a roll number, not a student name. A **foreign key** is a column that points at a primary key in another table, such as a result row that stores a roll number. **Referential integrity** means every foreign key either is empty, when the rules allow that, or matches a real row. You cannot record a result for a roll number that was never admitted.

Fact	A sensible type	Why not the neighbour
Roll number used only as an identifier	Text, or a whole number if it is truly numeric	If it has a leading zero, a number type will drop it
Marks	Number	Text cannot be averaged reliably
Fee paid or not	Yes/No (Boolean)	A text field of "yes" and "Y" and "paid" will not count
Date of admission	Date	Text dates sort alphabetically, so 01/12 comes before 02/01
Remarks	Long text	A short text field cuts the sentence off

Table 7. Data types you should be able to choose.

Relationships and the ER model

An **entity** is a thing we store: Student, Book, Subject. An **attribute** is a fact about it. A **relationship** is how entities belong together. In a relational database the relationship becomes a key that both tables share.

Degree	Meaning	Example	How it is stored
One-to-one	Each A matches at most one B	A student and that student's identity card	The card row holds the roll number, unique
One-to-many	One A matches many B; each B matches one A	One class, many students	The student row holds the class id
Many-to-many	Each side matches many of the other	Students and subjects	A third table of pairs, such as Enrolment

Table 8. The three relationship degrees, with a college example.

A **relational schema** is the written plan of the tables and keys, without the data. An **entity-relationship diagram** draws entities as rectangles, attributes as ovals or as a list inside the rectangle, and relationships as diamonds or as labelled lines. For a paper sketch, rectangles and labelled lines are enough if the degree is written on the line: 1 and M, or the words “one” and “many”.

WORKED EXAMPLE — A LIBRARY SCHEMA

Student(RollNo, Name, ClassName). Book(BookId, Title). Loan(LoanId, RollNo, BookId, DueDate).

RollNo is the primary key of Student. BookId is the primary key of Book. Loan.RollNo is a foreign key. Loan.BookId is a foreign key. One student can have many loans, and one book can appear in many loans over the year, so Student to Loan is one-to-many and Book to Loan is one-to-many. The many-to-many between students and books is carried by Loan, not by stuffing several book titles into one student cell. Referential integrity rejects a loan whose RollNo is not in Student.

COMMON MISTAKE

Repeating groups, such as Book1, Book2, Book3 as columns of Student, are not a relationship. They waste space, they cannot store a fourth book, and they make a query for “who has this title” miserable. Use another table.

Forms, queries, and reports

A **form** is a screen for entering or editing one row, with labels and checks, so the user never types into the raw table. A **query** is a question: which loans are due this week, which students have no loan, what is the average mark in Chemistry. A **criterion** is the condition, such as `DueDate <= today` or `Subject = "Chemistry"`. A **report** lays the answer out for printing, with a title, the rows, and a summary such as a count or a total. The same query can feed a report. Do not retype the numbers into a poster.

Summary statistics

A **summary statistic** compresses many values into one careful number. The **mean** is the sum divided by the count. The **median** is the middle value after sorting; if the count is even, average the two middle values. The **mode** is the most frequent value; a list can have more than one mode, or no single mode if every value appears once. The **range** is largest minus smallest. The mean moves when one extreme value is added. The median does not move as much. Say which one you used and why.

WORKED EXAMPLE — FIVE QUIZ SCORES

Scores: 10, 12, 12, 15, 41.

Mean = $(10+12+12+15+41) / 5 = 90 / 5 = 18$. Median = 12, the third value in the sorted list. Mode = 12. Range = $41 - 10 = 31$.

The mean is pulled up by 41. If the question is “what did a typical student score?”, the median describes the middle of this tiny class more honestly. If the question is “what was the class total worth?”, you need the sum, and the mean is the right average of that sum. The chart and the sentence must ask the same question.

Charts that match the question

A **bar chart** compares categories, such as mean mark by subject. A **pie chart** shows parts of one whole, and only when the parts add to that whole; too many slices make it unreadable. A **line graph** shows change over time, such as lab attendance by week. A **histogram** shows how a numeric measurement is spread, by grouping it into bins. A **scatter plot** shows two numeric variables together, one on each axis, so you can see whether they rise together. A **box plot** shows median and spread. Read a chart by naming it, reading both axes including the units, and only then describing the trend.

COMMON MISTAKE

A bar chart that does not start at zero can make a small difference look huge. A pie chart of average marks is the wrong picture, because those averages are not parts of one whole. Name the deception if a question asks you to criticise a chart.

A straight line, and what it does not prove

When a scatter plot looks roughly straight, a simple model is the line $y = mx + c$. Here x is the input you use to predict, y is the value you predict, m is the slope (how much y changes when x increases by 1), and c is the intercept (the predicted y when x is 0). A spreadsheet can fit the line and also report a **correlation** number. Correlation near 1 means the points rise together. Near -1 they move in opposite directions. Near 0 there is little straight-line relationship.

Correlation is not causation. Ice-cream sales and the use of fans rise together because summer drives both, not because ice-cream switches fans on. A model can still be useful for prediction: if study hours and quiz scores move together in your class data, the line is a summary of that association. It does not, by itself, prove that forcing extra hours will raise scores. Other factors — the test’s difficulty, who was absent — may be the real drivers. Those hidden factors are **confounding** factors.

WORKED EXAMPLE — READING A LINE

A fitted line says predicted quiz mark = $4 \times (\text{hours of practice}) + 20$.

The slope is 4: one extra hour goes with 4 extra marks in this dataset. The intercept is 20: the line predicts 20 marks at 0 hours. Predicting 3 hours gives $4 \times 3 + 20 = 32$. That is a prediction inside the range of the data, not a promise, and it is not a reason to claim the hours caused the marks. If nobody in the data practised 0 hours, do not make a speech about the intercept as a real student.

Population, sample, experiment

The **population** is everyone you want the conclusion to cover. A **sample** is the group you actually measure. A **parameter** is a number about the population, often unknown, such as the mean mark of every Class XI CS student in Karachi. A **statistic** is the matching number computed from the sample, such as the mean of your own section. A statistic estimates a parameter. It is not the parameter, and a sample of one section does not authorise a sentence that begins “Karachi

students are...”.

An **experiment** assigns the condition: this group uses the new worksheet, that group uses the old one, and the assignment is by chance. An **observational study** measures what people already do, without assigning it. A **survey** asks them. Only a careful experiment supports a cautious causal sentence, and even then only about the people and the conditions you studied. Observation can suggest a hypothesis. It should not be dressed up as proof.

Class XI stops at this idea of a model. You are not required to derive a formula for the slope by hand. You should be able to say what m and c mean, read a chart, and refuse a causal claim the data cannot carry. Building the line in a spreadsheet, and saying in one sentence what it does not prove, is the practical.

EXAM TIP

Database questions want the keys named on a specific scenario, not a definition copied in isolation. Statistics questions want the number and the choice: mean or median, and why. If both parts appear in one long question, use a heading for the schema and a heading for the chart.

Key terms

Term	Meaning in this lecture
DBMS	Software that stores related data and enforces rules.
Primary key	The column, or columns, that identify one row.
Foreign key	A column that must match a primary key in another table.
Referential integrity	No foreign key points at a missing row.
ER model	A drawing of entities, attributes, and relationships.
Query / criterion	A question, and the condition that filters the rows.
Mean / median / mode	Arithmetic average, middle of the sorted list, most frequent value.
Correlation	A measure of straight-line association, not a cause.
Population / sample	The whole group of interest, and the part you measured.
Parameter / statistic	A number about the population, and the estimate from the sample.
Experiment	A study where the condition is assigned, preferably at random.

Check your understanding

Attempt each question before you read the model answer. In the examination, a short question usually earns marks for a definition plus one precise point or a tiny example.

Why is a student name a poor primary key?

MODEL ANSWER

Names repeat, spellings change, and two students can share a name. A key must identify one row. A roll number is designed to do that.

Scores 4, 8, 8, 10. State the mode and the median.

MODEL ANSWER

The sorted list is already in order. Mode is 8. There are four values, so the median is the average of the second and third, $(8+8)/2 = 8$.

Your section's mean is 61. Is 61 a parameter?

MODEL ANSWER

No. It is a statistic about the sample you measured. It would estimate a parameter only if you had defined a wider population and treated the section as a sample of it — and one section is a weak sample of all BIEK candidates.

Board-style practice

Multiple choice

Each item is worth one mark. Cross out two options that cannot be right before you choose.

1. Referential integrity is broken when

- A. a foreign key matches no primary key
- B. a table has more than ten rows
- C. a query uses a criterion
- D. a report has a title

2. Students and subjects, where each student takes many subjects and each subject has many students, need

- A. one table with a subject column only
- B. a third table of pairs
- C. a pie chart
- D. two copies of the student name in the subject table

3. A line graph is the fitting chart for

- A. weekly attendance over a term
- B. the share of one fixed total among three clubs
- C. a list of phone numbers
- D. the primary key

4. Two variables rise together. The justified sentence is

- A. one must cause the other
- B. they are associated in this data
- C. the intercept is the cause
- D. the sample is the population

Answer key with reasons

Q	Answer	Why
1	A	The foreign key is supposed to point at a real row. A dangling reference breaks the rule.
2	B	Many-to-many is stored as a linking table, such as Enrolment(RollNo, SubjectCode).
3	A	A line is for change over time. A share of a whole is a pie, if the slices are few.
4	B	Association is what the scatter plot shows. Cause needs a design that can separate other explanations.

Short questions

S1. Define primary key and foreign key using the Loan and Student tables.

MODEL ANSWER

Student.RollNo is a primary key: it identifies one student and does not repeat. Loan.RollNo is a foreign key: it stores a roll number that must already exist in Student. Referential integrity is that rule.

S2. When is the median a better single summary than the mean?

MODEL ANSWER

When a few extreme values pull the mean away from the middle of the group. In 10, 12, 12, 15, 41 the mean is 18 and the median is 12. If the question is about a typical score, report the median and mention the extreme value rather than hiding it.

S3. Give one example of an experiment and one of an observation in a college.**MODEL ANSWER**

An experiment assigns, at random, one section to a new worksheet and another to the old worksheet, then compares quiz scores. An observation asks students how many hours they already study and compares those reports with scores, without assigning the hours. Only the first can support a careful sentence about the worksheet.

Long questions**L1. (5 marks) A clinic stores patients and appointments. A patient may have many appointments. An appointment is with one patient. Design the schema, name the keys, and state the relationship degree.****MODEL ANSWER**

Patient(PatientId, Name, Phone). Appointment(AppointmentId, PatientId, ApptDate, Reason).

PatientId is the primary key of Patient. AppointmentId is the primary key of Appointment.

Appointment.PatientId is a foreign key referencing Patient.PatientId. The relationship is one-to-many: one patient, many appointments, and each appointment belongs to one patient. Referential integrity forbids an appointment whose PatientId is not in Patient. Phone is not the primary key, because two patients can share a household number and a patient can change phones.

L2. (5 marks) Quiz marks in one section are 6, 7, 7, 9, 21. Compute the mean and the median. Recommend a chart if you must compare this section with two other sections, and write two sentences you are willing to say about a positive correlation between practice hours and these marks.**MODEL ANSWER**

Mean = $(6+7+7+9+21)/5 = 50/5 = 10$. Sorted, the middle value is 7, so the median is 7. The single 21 pulls the mean up; a comparison of “typical” performance should show the median as well as the mean.

A bar chart of the three section medians, with the axis starting at 0, compares categories. A pie chart does not.

A positive correlation would mean that, in this sample, higher practice hours go with higher marks along a rough straight line. It does not prove that practice caused the marks, because ability, the particular quiz, and who chose to practise are not controlled here. The honest second sentence limits the claim to this section’s data.

Practical work

The practical paper is 25 marks. Keep a journal: aim, steps, the program or diagram, the output, and one sentence on what went wrong the first time.

Lab 6 — Schema, query, and a chart

- 1 In the DBMS available in the lab (or on paper if the lab PCs are down), create Student and Result with a foreign key.
- 2 Enter four students and six results. Try to enter a result for a roll number that does not exist, and record the error.
- 3 Write a query for one subject and a report that shows the rows and the average.
- 4 In a spreadsheet, compute the mean and median of those marks and draw a bar chart of average by subject. State one thing the chart does not prove.

WHAT SUCCESS LOOKS LIKE

The illegal result is rejected. The average on the report matches a hand calculation. The written sentence refuses a causal claim.

Copy this summary

- One fact lives in one place. Primary keys identify rows. Foreign keys link tables. Referential integrity keeps the links real.
- One-to-many puts the key on the “many” side. Many-to-many needs a third table.
- Forms edit, queries ask, criteria filter, reports present.
- Mean is the arithmetic average. Median is the middle. Mode is the most common. Extremes pull the mean.
- Bars compare categories, lines show time, pies show a whole, scatter shows two numbers.
- $y = mx + c$ summarises a straight association. Correlation does not prove cause.
- A statistic describes a sample. A parameter describes a population. Experiments assign the condition; observations do not.

CLASS XI · UNIT 5

Lecture 5

Applications and Impacts of Computing

Textbook unit. Unit 5 — Applications and Impacts of Computing

Curriculum domain. E and F. Applications and impacts

Class time. about 30 periods

The same ideas from systems, programs, and data show up in devices that sense the world, in services that rent computing instead of buying it, and in records that many parties can check. This lecture treats the Internet of Things, cloud computing, and blockchain at the level of what makes them possible and where they help in Pakistan. It then turns to the human side: whose interests an automated decision serves, which sources deserve trust, and who is left out when connectivity grows.

Learning outcomes

- Describe the parts of an IoT system and the technologies that made IoT, cloud computing, and blockchain practical.
- Give a concrete use in a Pakistani setting without exaggerating what the technology guarantees.
- Identify stakeholders in an AI-style decision and a conflict between their interests.
- Separate a reliable information source from an unreliable one, and name bias in data.
- Discuss connectivity, assistive technology, the digital divide, and one environmental impact.

Internet of Things

The **Internet of Things (IoT)** is ordinary objects — meters, pumps, wristbands, classroom sensors — that collect data and communicate it so a person or another system can act. A typical chain has four parts. A **sensor** measures something physical, such as moisture or temperature. A **network** carries that reading. A **processor**, often helped by a **cloud** service, stores or judges it. An **actuator** does something in the world, such as opening a valve, or the “action” is simply a message to a person.

This became ordinary because several older limits moved at once: devices got smaller, they use less power, radios got cheap, and distant servers can store what a tiny device cannot. None of those is the IoT by itself. A thermometer with no link is just a thermometer.

Setting	What the system does	What it does not do
Smart city	Traffic or bin sensors report status to a dashboard	It does not replace road building or a maintenance budget
Industry	A machine reports vibration before it fails	A sensor does not repair the machine
Healthcare	A wearable shares a pulse trend with a clinic	It is not a diagnosis
Education	A lab logs equipment use	It does not teach the practical
Smart grid	Meters report use so supply can be planned	It does not create electricity
Smart farming	Soil sensors suggest when to irrigate	It does not know a pest it was not built to detect
Wearable	A band counts steps or heart rate	The reading can be wrong if the band is loose

Table 9. Uses named in the Class XI course, with a limit beside each so the answer stays honest.

WORKED EXAMPLE — A SINDH FARM, KEPT SMALL

A cotton field has three moisture sensors. Twice a day they send a number over the mobile network to a small cloud store. If the reading stays below a threshold the farmer chose, the farmer gets an SMS. The actuator is the farmer's own pump, not a fantasy robot.

The enabling technologies are a cheap sensor, a radio, and remote storage. The failure modes belong in the answer: no signal, a flat battery, a sensor in a puddle that is not typical of the field. A good design says who is allowed to see the farm's data. An open map of every field's water use may not be what the farmer agreed to.

Cloud computing

Cloud computing is renting storage, processing, or applications over the network instead of housing every server in the college. The college reaches them through the Internet. The appeal is scale: a result day needs more capacity than a Sunday, and the extra capacity can be temporary. The costs are a subscription, a dependence on the link, and a duty to know which country holds the data and who can be asked for it. "It is in the cloud" is not a security design. Access rules, encryption on the link, and backups still have to be chosen.

Blockchain, at the level this course needs

A **blockchain** is a shared ledger of records, grouped into blocks, linked so that changing an old record is obvious. Copies sit with more than one party. **Cryptography** — hashes and digital signatures — is what makes tampering visible and ties a record to the holder of a private key. The ledger does not need a single clerk whom everyone must trust to be the only copy.

That is useful when the problem is "we do not share one office, and we need to agree what was recorded": a degree verification hash that a university and an employer can both check, or a shipment log. It is a poor fit when the data changes every minute, when privacy requires that the data be deletable, or when a normal database with a responsible owner would do. A blockchain does not make a false document true. It makes a later silent edit harder. Putting a student's full record on a public ledger can itself be the harm.

EXAM TIP

A question that says "design an idea for Pakistan" wants the user, the data, the technology, and one risk. A definition of IoT with no scene, or a scene with no risk, is a thinner answer than one paragraph that has all four.

Stakeholders and automated decisions

An **artificial intelligence** system, even a simple one that ranks applications, has **stakeholders**: the people who gain or lose from its output. They rarely want the same thing. A college wants fast shortlisting. An applicant wants a fair chance and an explanation. A vendor wants the contract renewed. A parent wants to know the data will not be sold. Culture and values change what "fair" means: a form that assumes one kind of name, one language, or one kind of school will quietly disadvantage everyone else.

When you evaluate a design, name at least three stakeholders and one conflict. Then name a policy that reduces the harm: a right to ask for a human review, a limit on what is collected, a test of the system on more than one group before it is trusted, or a refusal to use the system for that decision at all. "The computer said so" is not an explanation.

Reliable and unreliable information

A **reliable source** can be checked. You can see who wrote it, when, and what evidence they used, and another careful source can be compared with it. An official board notification on the board's own site, a textbook passage, and a dataset with a described method are in that family. An **unreliable source** hides the author, uses urgency instead of evidence, or cannot be matched to anything else: a forwarded screenshot of a date sheet, an unnamed “expert”, a chart with no axis.

Bias enters before the chart. If a survey about college facilities is filled only by students who already have phones and data packages, the answers miss the students the question most concerns. If a face-detection demo is trained mostly on one kind of face, its errors will not be evenly spread. Humans are biased in what they bother to record. Computing does not wash that out. It can copy it quickly. Part of responsible use is to ask what was not collected.

Connectivity, people, and the environment

Computing raised **connectivity**. Email, the web, **Wi-Fi**, and **Bluetooth** let people and devices share data without a dedicated wire for every pair. Commerce uses that for orders and payments. Families use it to stay in touch. The same links carry harassment, scams, and rumours at the same speed as the useful messages. The impact is not “good” or “bad” as a single word. It depends on the design and on who can join.

Assistive technology is computing that reduces a barrier: a screen reader, captions on a video, a high-contrast display, voice input, a larger target on a touch screen. It is not a special favour added at the end. A portal that only works with a mouse, or a lecture video with no captions, excludes a student the college already enrolled. The same features help people on a noisy bus or in bright sun, so they are ordinary good design.

The **digital divide** is the gap in devices, connections, skills, language, and accessibility. A homework task that assumes a laptop and a quiet broadband evening is a different task in a household that shares one phone. Equity is not solved by a slogan. Practical steps include an offline copy of the worksheet, a lab hour, more than one language where the content allows, and not requiring a camera-on policy that some homes cannot meet.

The **environmental** impact is physical. Devices become **e-waste** when they are discarded with batteries and circuit boards still in them. Data centres use electricity, and so do blockchains that constantly recompute. The relieving side is also real: a form that removes a repeated rickshaw trip, a sensor that stops a pump watering a wet field. A balanced answer names one cost and one saving, and does not pretend the saving is automatic.

CULTURE AND DAILY LIFE

Connectivity changes how people coordinate family events, how shopkeepers take orders, and which language a young person reads most often. Those are cultural impacts. Mention one that you have actually seen, and keep it specific. A marker can tell a lived example from a slogan.

Key terms

Term	Meaning in this lecture
IoT	Objects that sense, communicate, and support an action.
Sensor / actuator	A device that measures, and a device that acts.
Cloud computing	Rented storage or processing reached over the network.
Blockchain	A shared, tamper-evident ledger held in more than one place.
Stakeholder	A person or group affected by a system's decisions.

Term	Meaning in this lecture
Reliable source	An account whose author, date, and evidence can be checked.
Bias	A systematic tilt in what was collected or how it is judged.
Assistive technology	Tools that reduce a barrier to using computing.
Digital divide	Unequal access to devices, links, skills, language, or accessibility.
E-waste	Discarded electronic equipment, including batteries.

Check your understanding

Attempt each question before you read the model answer. In the examination, a short question usually earns marks for a definition plus one precise point or a tiny example.

Name the four parts of a simple IoT chain.

MODEL ANSWER

A sensor, a network, processing or storage (often in the cloud), and an action — either an actuator or a message to a person.

Why is a public blockchain a bad place for a full medical file?

MODEL ANSWER

Many copies make deletion and privacy difficult, and the patient may not want every holder of the ledger to see the file. A hash used for checking a record is a different, smaller idea than publishing the record itself.

Give one check that a forwarded date-sheet image fails.

MODEL ANSWER

You cannot confirm the author or the date against the board's own notice. A screenshot can be edited. The reliable step is to open the board's site or the college's official notice.

Board-style practice

Multiple choice

Each item is worth one mark. Cross out two options that cannot be right before you choose.

1. Which set made IoT practical?

- A. Larger devices, no radios, and local-only files
- B. Smaller low-power devices, cheap connectivity, and remote computing
- C. Paper forms and a single clerk
- D. A faster printer

2. A blockchain's useful property in this course is that

- A. it makes false data true
- B. altering an old record is evident and copies are shared
- C. it replaces every database
- D. it uses no cryptography

3. A screen reader is best classified as

- A. e-waste
- B. assistive technology
- C. a network topology
- D. a primary key

4. A survey answered only by students with home broadband is a weak basis for a claim about all students because of

- A. referential integrity
- B. bias in who could respond
- C. bubble sort
- D. symmetric encryption

Answer key with reasons

Q	Answer	Why
1	B	The curriculum point is that size, power, connectivity, and processing moved together.
2	B	Tamper-evidence and a shared copy are the point. Cryptography is how that is achieved, so D is the opposite.
3	B	It reduces a barrier to using a computer.
4	B	The people who could not respond are exactly the people a digital-divide question is about.

Short questions

S1. Distinguish cloud computing from “saving a file on the PC in the lab”.

MODEL ANSWER

The lab PC keeps the file on local hardware the college owns and that others cannot reach once they leave. Cloud computing stores or processes the file on rented machines reached over the network, so it can be opened from elsewhere and scaled, and so it depends on the link and on the provider’s rules.

S2. Name three stakeholders in an automated college-admission shortlist and one conflict.

MODEL ANSWER

Applicants want a fair and explainable decision. The college wants speed and fewer clerks. The software vendor wants the product accepted. A conflict: the college’s wish for fully automatic rejection conflicts with the applicant’s wish for a human review when a certificate was scanned badly.

S3. State one environmental cost of computing and one way computing can reduce waste.

MODEL ANSWER

Discarded phones and lab PCs are e-waste if they are thrown out with their batteries. A moisture sensor that stops a pump on a wet field can save water and electricity. The saving happens only if someone acts on the reading.

Long questions

L1. (5 marks) Propose an IoT idea for a public school in Karachi during hot months. Include the sensor, the network, the action, the cloud’s role, and one privacy or reliability risk.

MODEL ANSWER

Each classroom has a temperature sensor. Readings go over the school’s Wi-Fi to a small cloud dashboard the head teacher can open. The action is not an automatic air conditioner the school may not have; it is a warning when a room stays above a chosen temperature, so the timetable can move that class, or a fan can be checked. The cloud stores the week’s readings so a single hot afternoon is not mistaken for a broken fan.

Reliability: if the Wi-Fi fails, the dashboard goes quiet, so the design needs a local display on the sensor itself. Privacy: the system measures rooms, not students’ bodies, and it does not need names. Adding cameras “while we are at it” would be a different, much harder decision and is not required for this idea.

L2. (5 marks) A video claims that a new vitamin “doubles marks”, and the only evidence is a chart with no axis and no author. Explain how you would judge the source, and connect your

judgement to bias and the digital divide.

MODEL ANSWER

The source fails the basic checks: no author, no date, no method, and a chart that cannot be read. I would look for a named study, a sample size, and a comparison that could be found again. A forwarded video is not that study.

Bias: the claim may come from an advertiser who selected only the students who improved. The digital divide matters because students who only meet the claim on a shared phone, in a language they are less sure of, have less chance to open the original source. Teaching people to ask for the author and the axis is part of equal access to information, not a separate moral lesson.

Practical work

The practical paper is 25 marks. Keep a journal: aim, steps, the program or diagram, the output, and one sentence on what went wrong the first time.

Lab 7 — One technology, one page

- 1 Choose IoT, cloud, or blockchain.
- 2 Describe a user in Pakistan, the data, where it is stored, and the action.
- 3 Name two stakeholders and one conflict.
- 4 Name one reliability risk and one privacy risk.
- 5 Find one source you would trust for a definition of the technology, and write why you trust it (author, organisation, date).

WHAT SUCCESS LOOKS LIKE

The page has a specific user and a specific risk. The source note says who published it and when, not only the website name.

Copy this summary

- IoT is sensor, network, processing, and an action. Small, low-power, connected devices plus the cloud made it practical.
- Name a Pakistani scene, and name what the system cannot do.
- Cloud computing rents capacity. It still needs access rules and a working link.
- A blockchain is a shared tamper-evident ledger. It does not make false data true, and it is a bad home for a secret file.
- List stakeholders and the conflict between them before you praise an automated decision.
- A reliable source has an author, a date, and evidence. Bias often begins with who was never measured.
- Assistive technology and the digital divide belong in the same answer as the benefits of Wi-Fi. E-waste is part of the environmental cost.

CLASS XI · UNIT 6

Lecture 6

Digital Literacy and a First Prototype

Textbook unit. Unit 6 — Digital Literacy

Curriculum domain. G and H. Digital literacy, and the Class XI prototype

Class time. about 20 periods

Digital literacy is the skill of finding, judging, collecting, and presenting information with digital tools. The Class XI textbook centres on advanced search and on designing a way to collect original data. The national outcomes also ask you to build and test a prototype of a small idea. That prototype closes this lecture, and Class XII will turn a surviving idea into a minimum viable product.

Learning outcomes

- Use advanced search operators and explain why the first result is not automatically the best.
- Write a research question and choose among interviews, surveys, prototypes, and simulations.
- Distinguish primary and secondary data and present findings in a fitting digital form.
- Design a small prototype, test it with a real user, and change one thing because of what you saw.

Advanced search

A search box is a tool with a grammar. A quoted phrase, "binary search", asks for those words together. A minus sign excludes a word: jaguar -car if you wanted the animal. site:.edu.pk limits the search to a kind of site. filetype:pdf limits the file type. OR keeps either term. These operators are how you locate a board notification, a dataset, or a definition without scrolling past advertisements.

The first link is a ranking, not a verdict. Rankings follow the search engine's formula and the advertiser's money. After you arrive, you still judge the page: who wrote it, which organisation they belong to, the date, and whether the evidence is on the page or merely asserted. Two sources that copy each other are not two sources. Prefer the organisation that owns the fact — the board for a date sheet, the statistics agency for a population figure.

WORKED EXAMPLE — FINDING A SYLLABUS FACT

You need the official name of the Class XI computer science book, not a shop's description.

Search "Computer Science" "Grade XI" site:.gov.pk and, separately, the Sindh Textbook Board's own pages. A shop page is useful as a clue to the chapter titles and must be labelled as a shop page if you cite it. The board or the directorate is the authority for what is prescribed. Write the date you looked, because pages change.

A research question worth collecting data for

A useful question says who, what, and the comparison or description you will actually make. "Is social media bad?" cannot be answered by a survey of your friends. "In our CS section, how many students used a paper trace before writing binary search, and did they finish the lab?" can be answered, and the answer is about that section only.

Primary data is data you collect for this question: your survey responses, your interview notes, your own measurements. **Secondary data** is data somebody else already published: a statistical yearbook, a previous practical's anonymous results, a dataset on a government site. Secondary data is not worse. It is often better, if you understand how it was collected. Say which kind you are using. Do not present a downloaded table as if you had surveyed the country.

Choosing a collection method

Method	You use it when	Watch out for
Survey	You need the same questions answered by many people	Vague wording; only the willing respond
Qualitative interview	You need the story behind a few people's choices	A small number cannot be written as a percentage of Pakistan
Prototype test	You need to see whether a design can be used	Polite classmates may say it is fine when it is not
Simulation	Trying the real system would be unsafe, slow, or impossible	A simulation is only as honest as its assumptions

Table 10. Methods named in the unit. Pick one main method and say why the others were worse for this question.

A survey question should be one question. “Do you like practicals and have a laptop?” cannot be answered with one yes. Offer a scale or clear options, and include “I don’t know” when that is a real answer. An interview needs a few open questions and the patience to write what was said, not what you hoped. A simulation, such as a paper model of a queue in the college canteen, should list the assumptions: how many servers, how long an order takes.

Collect only what the question needs. Names are usually unnecessary on a class survey about study habits; a tick is enough. Tell people what the data is for, that they can refuse, and that you will not pass the sheet to anyone outside the practical. That is the minimum of respectful data collection. It is also the habit behind the privacy talk in the previous lecture.

Presenting the result

The form follows the audience. A teacher marking a practical can read a one-page report: question, method, table, chart, and a limit on the claim. A class noticeboard might need an **infographic** with one chart and one sentence. A short slide deck fits a three-minute presentation. A spreadsheet holds the workings so the chart can be checked. Do not paste a spreadsheet onto a slide.

Every chart needs a title, labelled axes or a legend, and the source and sample size nearby: “n = 32 students in Section A, 8 October 2026.” A sentence under the chart should say what a careful reader may conclude. “Most of this section starts the practical on paper” is allowed if the numbers say so. “Pakistani students prefer paper” is not.

WORKED EXAMPLE — A ONE-PAGE FINDING

Question: in our section, did students who said they traced binary search on paper also finish the lab task?

Method: an anonymous survey of one section, four questions, done at the end of the lab. Primary data. n = 28 of 32 present.

Result: 18 said they traced on paper; 16 of those finished. 10 said they did not; 4 of those finished. A grouped bar chart shows finished and not finished for the two groups.

Limit: people may misremember, the section is not all of BIEK, and finishing the lab has other causes. The prototype of a “trace-first checklist” can be tested next, which is a different question.

A prototype of a small idea

A **prototype** is a cheap, early version of an idea, built so you can learn. It can be a paper screen, a slide that pretends to be an app, or a script that does one step. It is not the finished product and it is not a business yet. You **test** it by watching somebody who was not the designer try to use it, without you explaining. You **iterate** by changing the prototype because of what you saw, then testing again.

Pick a problem you have actually met. A one-page “which search do I use?” card for Class XI, a paper timetable of the lab PCs, or a three-screen sketch of a lost-and-found board are all large enough. A plan to “disrupt education in Pakistan” is not a prototype. It has no user you can sit next to this week.

WORKED EXAMPLE — PAPER PROTOTYPE OF A LAB CHECKLIST

Problem: students skip the trace and then cannot explain a wrong index.

Prototype: a paper card with three boxes — list, target, low/high/middle — and a final line “index or not found”.

Test: two classmates use it on a new list while you stay quiet. One of them does not know where to write the middle value. Iteration: add the words “middle value” under the third box and test with a third classmate.

What you learned is about the card, not about the whole country. Class XII will ask which assumption is the riskiest if you ever try to offer the card as a product.

COMMON MISTAKE

Do not paste pages from the textbook or from another author’s notes into a prototype you claim as yours. A prototype can show your own examples, your own layout, and your own questions. Copying a book is not design, and it is not allowed as practical work.

EXAM TIP

Digital-literacy questions reward a method matched to a question, a labelled chart, and a sentence that limits the claim. Prototype questions reward a user, what you built, what you saw in the test, and what you changed. A definition of “survey” with no question attached is the weak version of this answer.

Key terms

Term	Meaning in this lecture
Search operator	Grammar in the search box, such as quotes, minus, site:, and filetype:.
Primary data	Data you collect to answer this question.
Secondary data	Data already collected by someone else.
Survey	The same questions asked of many people.
Qualitative interview	A conversation that collects an account, not only a tick.
Simulation	A model of a system you cannot safely or cheaply run for real.
Infographic	A visual summary aimed at a quick reading.
Prototype	An early, cheap version built so a test can teach you.
Iteration	A change made because of a test, followed by another test.

Check your understanding

Attempt each question before you read the model answer. In the examination, a short question usually earns marks for a definition plus one precise point or a tiny example.

What does the search `filetype:pdf site:.edu.pk binary search` **try to do?**

MODEL ANSWER

It looks for PDF files on .edu.pk sites that mention binary search. It does not prove that those files are correct or current.

Why is “n = 28 in Section A” required next to a chart?

MODEL ANSWER

The reader must know who was measured. Without it, a bar can be mistaken for a national fact.

What is the difference between a prototype and a finished app?**MODEL ANSWER**

A prototype is built to learn, often on paper or as a thin sketch, and it is expected to change after a test. A finished app is built to be used for real and maintained. This year you are marked on the learning, not on a polished product.

Board-style practice**Multiple choice**

Each item is worth one mark. Cross out two options that cannot be right before you choose.

1. Quotes around a phrase in a search box

- A. exclude that phrase
- B. ask for those words together
- C. limit the search to books you own
- D. prove the page is official

2. Data you download from a statistics agency is

- A. primary data for you
- B. secondary data
- C. a prototype
- D. a primary key

3. The best reason to interview three students instead of surveying them is that you want

- A. a percentage for the whole city
- B. a detailed account of how they study
- C. a pie chart with three slices
- D. to avoid writing anything down

4. After a prototype test you should

- A. hide the problems so the marks stay high
- B. change the prototype because of what the user did
- C. add every feature you can imagine
- D. throw away the notes

Answer key with reasons

Q	Answer	Why
1	B	Quotes mean the words should appear as a phrase. They say nothing about whether the page is trustworthy.
2	B	Somebody else collected it. It becomes part of your study as secondary data, clearly cited.
3	B	Interviews collect depth. Three people are not a percentage of a city.
4	B	Iteration is the point of the test. The notes are the evidence.

Short questions

S1. Write one research question about your own CS class that a survey can answer, and one claim that the survey must not make.

MODEL ANSWER

Question: of the students present in this section today, how many have used the college lab printer this month? The survey must not claim that “BIEK students lack printers”, because one section on one day is not the board and the question did not ask about home printers.

S2. Give two rules for an honest chart in a practical report.**MODEL ANSWER**

Label the axes or the slices, and start a bar chart’s quantity axis at zero unless you have a reason you are willing to explain. State the sample size and who was included. The title should describe the chart, not shout a conclusion the data does not carry.

S3. Describe a prototype test in three steps.**MODEL ANSWER**

Give the prototype to a user who did not design it. Set a task and stay quiet. Write what they did and where they stopped, then change one part of the prototype and, if there is time, test that change.

Long questions

L1. (5 marks) You want to know how Class XI CS students in your college find past papers. Design the data collection. Include the question, the method, primary or secondary data, one ethical point, and how you would present the result.

MODEL ANSWER

Research question: in our college, which sources do Class XI CS students report using when they look for past papers, and do they check the board site?

Method: a short anonymous survey of the CS sections, with options such as board site, teacher, shop notes, and forwarded messages, plus a yes/no on checking the official site. This is primary data. A secondary source, such as the board site itself, can be described separately so you know what the official source is, but it does not answer what students do.

Ethics: no names, a sentence at the top saying the practical is the only use, and the right to leave it blank.

Presentation: a bar chart of the sources, n written under it, and a sentence limited to the students who responded. A slide with that one chart is enough for the class. The spreadsheet stays in the journal so the bars can be checked.

L2. (5 marks) Design a prototype that helps a classmate choose between linear search and binary search. Describe the test and one iteration.

MODEL ANSWER

The user is a Class XI student who has met both algorithms. The prototype is a paper card. Line 1 asks: is the list sorted? If no, it points to linear search. If yes, it points to binary search and reminds the user to write low, high, and middle. A tiny example list sits at the bottom, written by the designer, not copied from a book.

Test: a classmate is given an unsorted list and asked to choose an algorithm using only the card. Suppose they still try binary search. The iteration adds the words “Stop. Do not use binary search on an unsorted list” on the No branch, in larger writing, and the card is tested with another person.

Success is not that the card is pretty. Success is that the second user takes the correct branch without an oral hint.

Practical work

The practical paper is 25 marks. Keep a journal: aim, steps, the program or diagram, the output, and one sentence on what went wrong the first time.

Lab 8 — Search, survey, and a prototype

- 1 Write down two different searches you used, including at least one operator, and what you decided about the pages you opened.
- 2 Run a five-question anonymous survey in your own section, or analyse a data table your teacher provides and label it secondary.
- 3 Produce one chart with a title, a sample size, and a two-sentence limit on the claim.
- 4 Make a paper prototype of one study aid. Test it with one classmate. Change one thing. Photograph or redraw both versions for the journal.

WHAT SUCCESS LOOKS LIKE

The journal shows the search strings, the chart with n, and both versions of the prototype with a sentence on what the user did.

Copy this summary

- Quotes, minus, site:, and filetype: narrow a search. They do not certify the page.
- Judge the author, the date, and the evidence. Two copies of the same rumour are one source.
- A research question names the group and the fact you will actually collect.
- Primary data is yours. Secondary data is cited as someone else's.
- Surveys count the same questions. Interviews collect accounts. Prototypes test a design. Simulations stand in when the real run is impossible.
- Charts need labels and a sample size. The sentence under the chart must fit the sample.
- A prototype is for learning. Watch a user, change one thing, and keep the notes.

Revision and the practical journal

Formulas and traces worth memorising

Item	Keep this exact
NOT	01 and 10
AND / OR	AND is 1 only if all inputs are 1. OR is 0 only if all inputs are 0.
NAND	Opposite of AND. Universal.
XOR	1 when the inputs differ. $A \text{ XOR } B = A'B + AB'$.
De Morgan	$(A+B)' = A'B'$ and $(A \cdot B)' = A' + B'$.
Duality	Swap AND with OR and 0 with 1. Variables stay.
Binary search	Sorted list only. Middle by integer division. Move low or high. Stop when found or low passes high.
Bubble sort	Swap unordered neighbours. Show each pass.
Insertion sort	Grow a sorted portion on the left.
Mean	Sum divided by count. Median is the middle of the sorted list.
Line	$y = mx + c$. Correlation is not causation.
List index	Python starts at 0. Board pseudocode in these notes starts at 1 unless a question says otherwise.

Practical journal for the 25 marks

- Lab 1: an eight-row truth table that matches a circuit.
- Lab 2: an SDLC page with three black-box tests.
- Lab 3: a binary-search trace, including a missing item.
- Lab 4: a tested function and a list search, with a dry run.
- Lab 5: a turtle compound shape and the angle you chose.
- Lab 6: a schema that rejects a bad foreign key, plus a chart with a refused causal claim.
- Lab 7: one technology, two stakeholders, one source you trust, and why.
- Lab 8: a search string, a chart with n , and two versions of a prototype.

Take the journal to the practical. A marker should be able to repeat one test from your notes and get your result.